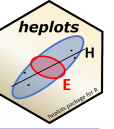


Visualizing Multivariate Linear Models with heplots:: CHEATSHEET



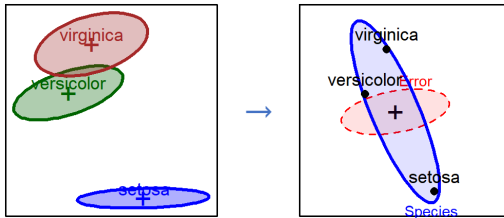
① Fit & visualize

fit an MLM, then see its tests as H and E ellipses

Basics

A **multivariate linear model** `lm(cbind(y1, y2, ...) ~ x1 + x2 + ...)` gives an object of class "mlm". Tests of each term use two **SSP matrices**: **H** (hypothesis) and **E** (error).

An **HE plot** shows **H** and **E** as ellipses (ellipsoids in 3D) in the space of two (or three) responses.



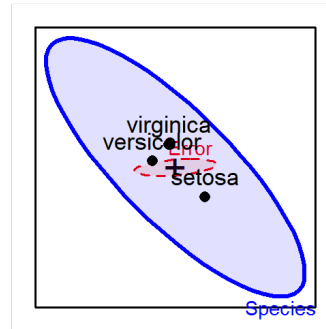
data ellipses, by group

HE plot (effect scaling)

- **Significance scaling** (default, `size = "evidence"`): if H sticks out beyond E **anywhere in a plot**, the term is significant by Roy's test ($\alpha = .05$). But H inside E in one view does **not** mean "not significant": the effect may lie in other directions (try `pairs()` or `candisc()`).
- `size = "effect"`: H at its actual size relative to E (effect size). E, the residual covariance, is the same in both; only H is rescaled.
- The **orientation** of H shows *how* the responses relate to the effect.

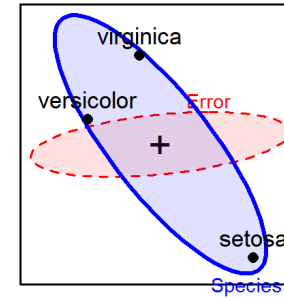
HE plot family

`mod <- lm(cbind(Sepal.Length, ..., Petal.Width) ~ Species, data = iris)`



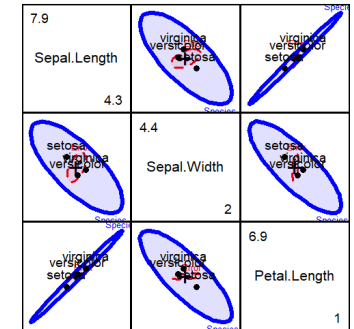
`heplot(mod)`

2D HE plot for two responses



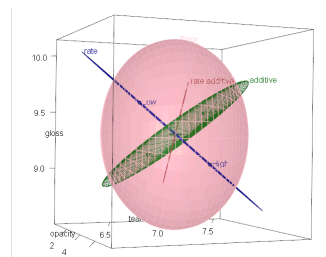
`heplot(mod, size = "effect")`

effect scaling: same E; H not inflated, so it shows effect size



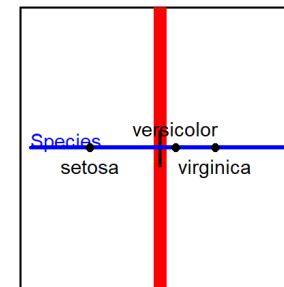
`pairs(mod)`

all pairwise HE plots: an HE plot matrix



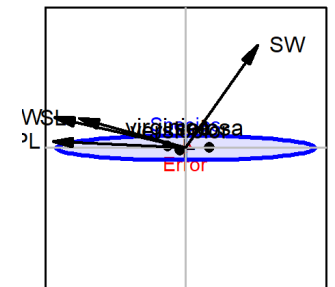
`heplot3d(mod)`

3D HE ellipsoids (rgl); spin & zoom



`heplot1d(mod)`

1D HE plot: H & E as intervals



`heplot(candisc(mod))`

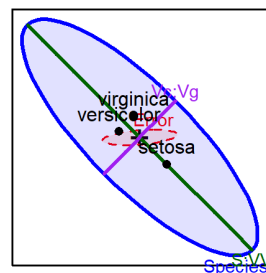
candisc: HE plot in canonical space, with variable vectors

Complete the template

```
mod <- lm(cbind(<Y1>, <Y2>, ...)
~ <TERMS>, data = <DATA>)
car::Anova(mod) # MANOVA tests
heplot(mod,
variables = <1:2>, # responses
terms = <TERMS>, # H ellipses
hypotheses = <HYP>, # contrasts
size = "evidence", # "effect"
fill = TRUE)
```

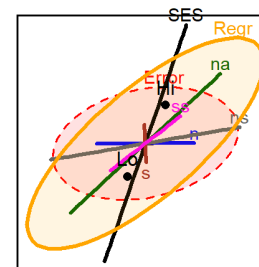
Tests of linear hypotheses

hypotheses =: contrasts or linear combinations of coefficients



`heplot(mod, hypotheses = hyp)`

1-df contrasts are drawn as **lines**: e.g., setosa vs. others; versicolor vs. virginica



`heplot(mod, hypotheses = list(Regr = preds))`
MMRA: overall test of all predictors vs. each one (Rohwer)

A 1-df H "ellipse" is a line. For a quantitative predictor, it shows the direction of its regression coefficients in the response space.
A joint hypothesis (Regr) collects several terms into one H ellipse.

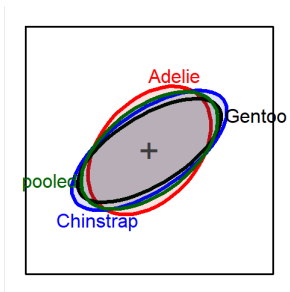
Visualizing Multivariate Linear Models with heplots :: CHEATSHEET



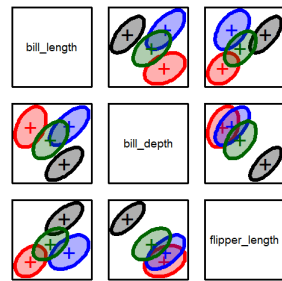
② Check, refine & report

assumptions, customization, statistics

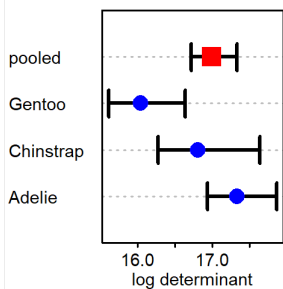
Homogeneity of covariance *peng data*



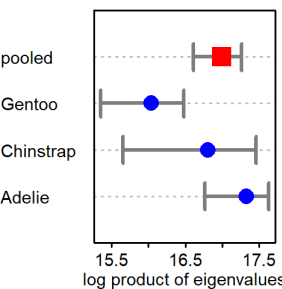
`covEllipses(Y, group)`
within-group covariance ellipses (+ pooled)



`covEllipses(..., variables = 1:3)`
scatterplot matrix version



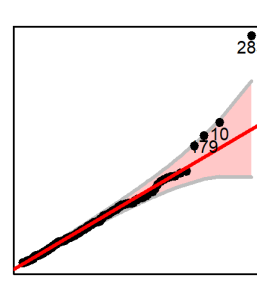
`plot(boxM(mod))`
log determinants of group covariance matrices, with CIs



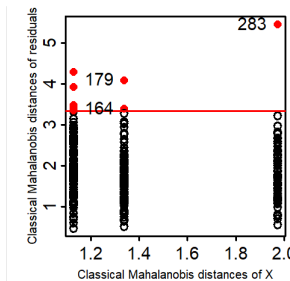
`plot_boxM_boot(...)`
the same, with bootstrap CIs; also for other eigenvalue statistics

`boxM(Y, group)` or `boxM(mod)`: Box's M test.
`logdetCI()`, `eigstatCI()`, `traceCI()`: other summaries.
`bartlettTests()`, `leveneTests()`: univariate tests, one per response.

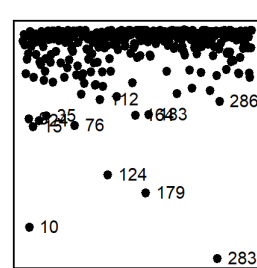
Normality, outliers & robust fits *peng data*



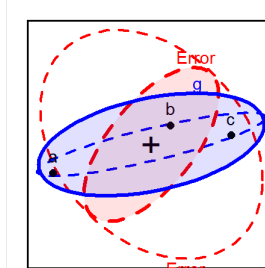
`cqplot(mod)`
 χ^2 QQ plot of squared Mahalanobis distances of residuals



`distancePlot(mod)`
distances of X vs. residuals, with cutoff; labeled outliers



`plot(robmlm(mod))`
observation weights from a robust MLM

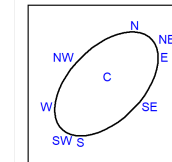


`heplot(rmod)`
`heplot(mod, add = TRUE)`
robust (solid) vs. classical (dashed): outliers inflate classical E (simulated)

`Mahalanobis()`: with robust center/cov. `noteworthy()`: unusual points to label. See **mvinfluence** for influence diagnostics.

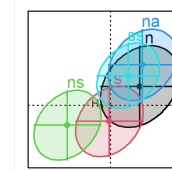
Customizing HE plots

- variables: which responses (indices or names)
- terms, hypotheses: what to draw; `term.labels`, `hyp.labels`
- size = "evidence" | "effect", `alpha`, level
- col, lty, lwd, fill, fill.alpha: one per ellipse
- factor.means, grand.mean, `markH0`, `center.pch`
- add = TRUE: overlay; `xlim`, `ylim`, `offset.axes`
- Repeated measures: `idata`, `idesign`, `imatrix`, `iterm`



`label.pos`: where to put labels
• 0:4 = C, S, W, N, E
• compass "N", "NE", ... "NW"
• a fraction 0–1 around the ellipse
• see `label.ellipse()`

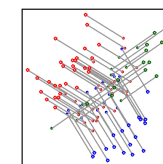
Coefficients *Rohwer data*



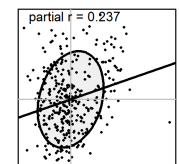
`coefplot(mod)`
joint 95% confidence ellipses for each predictor's coefficients on two responses; excluding (0, 0) $\Rightarrow$ significant for those two responses jointly (unadjusted)

`stdcoef(mod)`, `stdmodel(mod)`: standardized coefficients & model

Other plots *peng data*



`interpPlot(xy1, xy2)`
interpolate between two data sets, e.g., data $\rightarrow$ residuals



`pvPlot(Y, vars)`
partial variables plot: two variables, controlling for the others

Model statistics

Function	Returns
<code>car::Anova(mod)</code>	multivariate tests: Pillai, Wilks, H-L, Roy
<code>etasq(mod)</code>	multivariate η^2 (partial effect size) per term
<code>uniStats(mod)</code>	univariate R^2 , F, p for each response
<code>glance(mod)</code>	one row per response, like <code>broom::glance()</code>

Function	Returns
<code>termMeans(mod, "term")</code>	response means for the levels of a factor
<code>vcov(mod)</code>	covariance matrix of the coefficients
<code>df.terms(mod)</code>	df for each term
<code>colMeansList()</code> , <code>covList()</code>	statistics for each group

Learn more

- **Vignettes:** `HE_manova`, `HE_mmra`, `Robust`, `datasets`
- **Related:** `candisc` (canonical views), **`mvinfluence`**, `car`, `rgl`
- Friendly (2007) *JCGS*; Friendly, Monette & Fox (2013) *Stat. Sci.*

Datasets (37; see `vignette("datasets")`):

MANOVA Plastic, Skulls, AddHealth, MockJury, peng
MMRA Rohwer, Hernior, SocGrades, NLSY
Repeated VocabGrowth, WeightLoss, Adopted